

Probing Functional Correctness in Diffusion Language Models

Guan-Ming Chiu

National Taiwan University
gmchiu@arbor.ee.ntu.edu.tw

Jeng-Yue Liu

Carnegie Mellon University
buffett1@andrew.cmu.edu

Abstract

Diffusion language models generate text by iteratively denoising all tokens in parallel, but when and where their hidden states encode whether the output will be functionally correct remains unknown. We present the first probing study of DLM internals, training linear classifiers on hidden states to predict functional correctness. Across two models (LLaDA-8B, Dream-7B) and four tasks, we find that DLMs uniquely accumulate correctness signal across denoising steps (AUC gains of 0.08–0.11 on reasoning tasks), absent in single-pass AR decoding. However, step-0 signal reflects prompt difficulty rather than diffusion-specific computation. Signal emergence is task-dependent: structural tasks show flat profiles while reasoning tasks show gradual buildup. The two models exhibit distinct layer dynamics, with LLaDA concentrating signal in upper layers while Dream redistributes toward lower layers. We further show that probe confidence can identify likely failures, enabling selective generation that avoids 36–98% of wasted compute.

1 Introduction

Diffusion language models (DLMs) have emerged as a promising alternative to autoregressive (AR) generation (Austin et al., 2021a; Sahoo et al., 2024; Lou et al., 2024). Unlike AR models, which generate tokens sequentially, DLMs produce all output tokens simultaneously through iterative denoising: starting from a fully masked or noisy sequence and progressively refining it over many steps.

This parallel generation process raises a fundamental question: *when* during denoising, and *where* within the model’s layers, does the network encode whether its output will be functionally correct? In AR models, probing classifiers have revealed how linguistic and factual knowledge localizes across layers and positions (Tenney et al., 2019; Geva et al., 2023; Azaria and Mitchell, 2023). However, no analogous study exists for diffusion language

models, whose bidirectional, iterative generation process creates a fundamentally different representational landscape.

We address this gap with the first probing study of DLM hidden states. Our contributions are: (1) DLMs uniquely accumulate correctness signal across iterative denoising, a capacity absent in single-pass AR decoding: while step-0 hidden states already carry above-chance signal comparable to an AR probe on the same prompt (and thus largely attributable to prompt difficulty rather than diffusion), DLMs gain a further 0.08–0.11 AUC through subsequent denoising on reasoning tasks; (2) we identify task-dependent emergence patterns, with structural tasks showing flat AUC profiles from step 0 and reasoning/code tasks requiring iterative refinement; and (3) we observe distinct layer dynamics between LLaDA and Dream, with LLaDA concentrating signal in upper layers while Dream redistributes signal toward lower layers on simple tasks. Code and data are available at <https://github.com/guan404ming/dllm-probing>.

2 Background

Diffusion language models. LLaDA (Nie et al., 2025) uses block masking during training and generates by progressively unmasking tokens over a fixed schedule. Dream (Ye et al., 2025) employs a global masking schedule with a different unmasking strategy. Both models condition on a fully visible prompt and iteratively denoise a generation region of masked tokens.

Probing classifiers. Probing involves training simple classifiers on frozen model representations to test whether specific information is encoded (Belingkov, 2022; Conneau et al., 2018). We adopt this methodology for DLMs, probing hidden states at intermediate denoising steps.

Related work. Prior studies have probed AR model representations for linguistic structure (Tenney et al., 2019), factual knowledge (Geva et al., 2023), and truthfulness (Azaria and Mitchell, 2023). Confidence-based methods for AR models use early-exit thresholds to skip computation (Schuster et al., 2022; Elbayad et al., 2020). Two concurrent works study DLM representations from complementary angles. Shnaidman et al. (2025) demonstrate activation steering for safety behaviors in masked DLMs, showing that DLM representations are amenable to linear intervention. Wang et al. (2025) identify temporal oscillation in DLM outputs, where correct answers appear mid-denoising but are overwritten in later steps. Mündler et al. (2025) propose constrained decoding with context-free grammars to enforce syntactic correctness in DLM outputs; our work instead asks whether the model’s own representations already encode whether the output will be correct, without external constraints. We complement these prior findings by probing hidden representations for functional correctness across the full layer-step grid, characterizing *when* and *where* correctness signal emerges.

3 Method

Hidden state extraction. Given a DLM with L layers, we run 128-step denoising and extract hidden states at 7 checkpoint steps $t \in \{0, 1, 4, 16, 32, 64, 127\}$. At each checkpoint, we obtain the full hidden state tensor of shape $(L, \text{gen_length}, d)$. We partition the generation region into 4 equal-length position regions and mean-pool within each region, yielding features of shape $(L, 4, d)$ per instance per step. We choose 4 regions to align with LLaDA’s block length of 32 tokens ($\text{gen_length} / \text{block_length} = 4$ for JSON schema’s 128-token blocks) while keeping the number of features manageable given the limited sample sizes. An ablation over 1, 2, and 4 regions shows best AUC varies by less than 0.02 across configurations (Appendix G).

Probe training. For each (step, layer, region) triple, we reduce dimensionality with PCA to 64 components, apply standard scaling, and train a logistic regression classifier to predict functional correctness (Pedregosa et al., 2011). We evaluate with 5-fold stratified cross-validation, reporting AUC as our primary metric. Control probes trained on shuffled labels (3 random permutations, max

AUC across the same layer-region grid) yield 0.53–0.58 across all eight (dataset, model) settings, well below the actual probe AUCs of 0.65–0.88 (dashed grey trace in Figure 2); the slight offset above 0.50 reflects multiple-comparison selection across layers and folds, and is uniform across configurations.

Functional correctness. For JSON schema validation, an output is correct if it parses as valid JSON and matches the reference structure. For math reasoning (GSM8K), an output is correct if the extracted numeric answer matches the ground truth. For code generation (MBPP), an output is correct if the generated function passes all provided test assertions. For science QA (ARC), an output is correct if the extracted answer letter matches the gold label.

Selective generation. We simulate instance-level filtering by training a per-step probe (using the best layer and region from the full analysis). Let p be the probe’s predicted probability of functional correctness and define confidence as $\max(p, 1 - p)$. At inference time, if the probe predicts $p < 0.5$ with confidence exceeding a threshold τ , we skip the remaining denoising steps for that instance entirely, producing no output rather than wasting compute on a generation predicted to be incorrect. We report the fraction of instances filtered and the probe’s classification accuracy.

4 Experimental Setup

Models. We evaluate LLaDA-8B-Instruct (33 layers) (Nie et al., 2025) and Dream-7B-Instruct (29 layers) (Ye et al., 2025), two open-weight DLMs with different masking strategies.

Datasets. We select four tasks spanning structural to reasoning-heavy complexity, testing whether iterative denoising contributes differently by task type: (1) **JSON schema validation** (272 instances from eth-sri/json-mode-eval-extended, an extension of NousResearch/json-mode-eval that augments the original 100-instance benchmark with additional schema variants); each instance pairs a JSON Schema with a natural-language request, and a generation is functional if it parses as JSON and matches the reference structure exactly; (2) **GSM8K** (1,319 test instances) (Cobbe et al., 2021), multi-step math reasoning; (3) **MBPP** (257 sanitized test instances) (Austin et al., 2021b), code generation verified by execution; and (4)

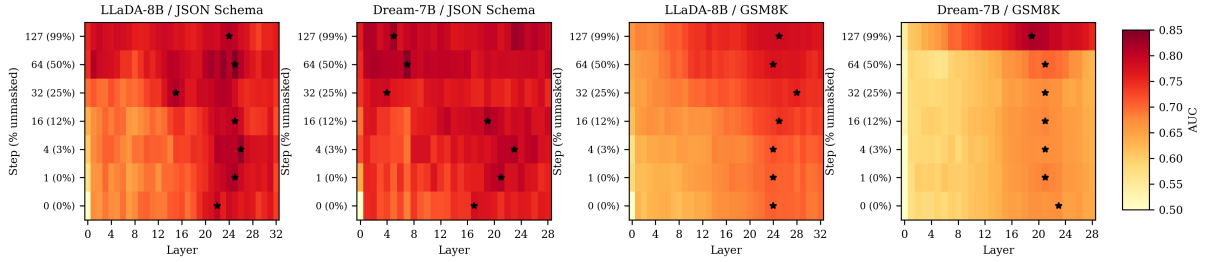


Figure 1: AUC heatmaps (layer $\times$ step) for JSON schema and GSM8K. Each cell shows the best AUC across the 4 position regions for that (layer, step) pair. Stars mark the best layer per step. JSON schema shows strong signal from step 0; GSM8K shows gradual emergence. Note Dream’s layer migration on JSON schema. Per-cell standard deviations across 5-fold CV are ≤ 0.05 . MBPP and ARC heatmaps in Appendix B.

	JSON	GSM8K	MBPP	ARC
LLaDA-8B	48.5%	66.3%	40.9%	78.2%
Dream-7B	46.0%	61.5%	44.4%	74.5%

Table 1: Baseline functional correctness rates (seed=0, 128 denoising steps).

ARC-Challenge (1,172 test instances) (Clark et al., 2018), multiple-choice science QA with short answers that isolate reasoning from generation length. Generation lengths are 256, 512, 256, and 256 tokens, respectively.

Baseline functional rates. Table 1 shows the functional correctness rates with 128 denoising steps. All datasets have reasonable class balance, enabling meaningful probe training.

5 Results

5.1 Task-Dependent Emergence

Figure 1 shows AUC heatmaps for JSON schema and GSM8K (MBPP and ARC in Appendix B). Two distinct emergence patterns are visible.

For **JSON schema**, correctness signal is already strong at step 0 (AUC $\approx 0.78\text{--}0.80 \pm 0.04$) and remains relatively flat across all denoising steps. This indicates that the prompt encoding alone, before any denoising occurs, already contains signal predictive of structural correctness.

For **GSM8K**, step-0 signal is already well above chance (AUC $\approx 0.67\text{--}0.71 \pm 0.03$), but denoising brings substantially more gain: AUC rises to $\approx 0.78\text{--}0.82 \pm 0.03$ by step 127 (Figure 2).

MBPP follows the same gradual-buildup pattern as GSM8K: step-0 AUC is $\approx 0.75\text{--}0.76 \pm 0.04$ and peaks at $\approx 0.84\text{--}0.87 \pm 0.05$ at later steps (step 64 for LLaDA, step 127 for Dream).

ARC-Challenge shows the weakest and most

denoising-dependent signal: step-0 AUC is $\approx 0.61\text{--}0.67$, and signal builds gradually to $\approx 0.71\text{--}0.75$ by step 64–127. Dream’s ARC signal is particularly flat until a sharp jump at the final step ($0.61 \rightarrow 0.71$), suggesting science reasoning requires nearly complete denoising before correctness is predictable.

All four tasks have above-chance step-0 signal, but JSON schema gains only $\Delta\text{AUC} \approx 0.04$ from denoising while GSM8K, MBPP, and ARC gain $\approx 0.08\text{--}0.11$, reflecting the difference between structural and reasoning tasks. Standard deviations across 5-fold CV are ≤ 0.05 for all configurations, indicating stable estimates. Because DLMs process all generation positions jointly via bidirectional attention, correctness-relevant information is available across the full output region from step 0.

We also tested whether the probe could select better random seeds: scoring 5 seeds per instance at step 1 and running full denoising only for the top-scoring seed yields +0.0% improvement for both models on JSON schema, confirming that the step-0/1 signal reflects instance-level difficulty rather than seed-specific trajectory quality.

5.2 Divergent Layer Dynamics

The heatmaps in Figure 1 reveal distinct layer dynamics between LLaDA and Dream (full per-step best layers in Table 3, Appendix C).

LLaDA concentrates its best probing signal consistently in upper layers (L22–28) across JSON schema, GSM8K, and ARC, similar to patterns observed in AR transformers. MBPP is an exception, where best layers are more variable (L15–31).

Dream shows a task-dependent migration pattern. On JSON schema, the best layer migrates from the upper range (L17–23 at steps 0–16) to lower layers (L4–7 at steps 32–127). On GSM8K,

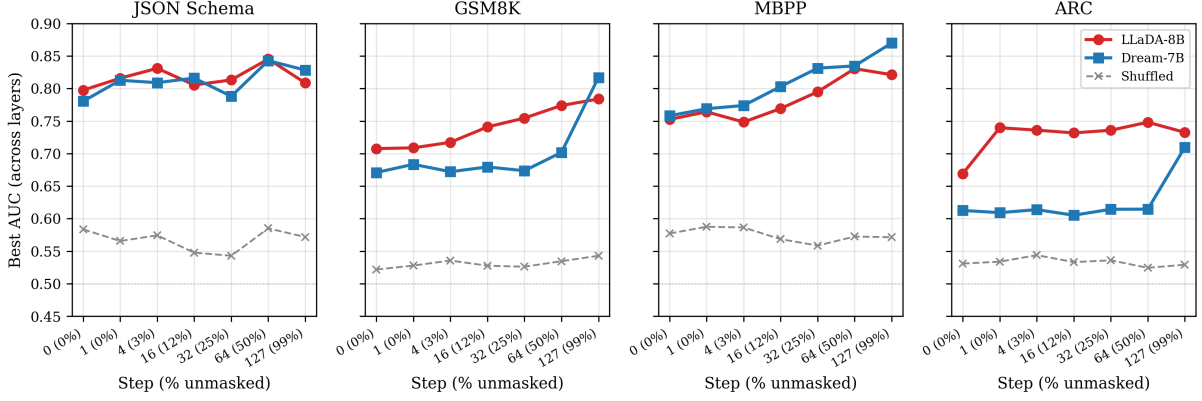


Figure 2: Best AUC across layers at each denoising step (x-axis annotated with cumulative % of generation tokens unmasked). JSON schema is flat (≈ 0.80 from step 0), while GSM8K, MBPP, and ARC show gradual emergence. Dashed grey trace is the shuffled-label control (mean AUC across both models, averaged over 3 random label permutations); the dotted line marks chance (0.50). Standard deviations across 5-fold CV are ≤ 0.05 for all probe points.

MBPP, and ARC, however, Dream’s best layers remain in the mid-to-upper range (L15–25) throughout denoising, with no downward migration. We discuss a possible interpretation in terms of Dream’s AR pretraining origin in §6.

5.3 Filtering Hopeless Instances (Offline)

We evaluate whether per-step probe confidence can identify instances whose outputs will be incorrect, enabling a hypothetical system to skip generation entirely for hopeless instances. This is an *offline* simulation on hidden states already extracted from full 128-step generation, intended to characterize the upper bound of what the probe can offer; it does not account for the cost of running the probe in a live inference loop. Table 2 shows results with a confidence threshold of $\tau = 0.80$.

The results are strongly task-dependent. For JSON schema and MBPP, both models save over 95% of compute and clear the majority baseline by 7–22 percentage points: the probe is confident from step 0, identifying most instances predicted to fail immediately. For GSM8K and ARC, savings are more modest (36–83%) because reasoning tasks require more denoising steps before the probe can confidently distinguish hopeless instances. For ARC, where the baseline functional rate exceeds 74% (Table 1), classification accuracy is only marginally above the majority baseline, indicating limited discriminative value on this task.

Comparison to confidence heuristics. A natural concern is whether a hidden-state probe is necessary at all: the model already exposes a per-token

	JSON	GSM8K	MBPP	ARC
<i>LLaDA-8B</i>				
Compute saved	98.4%	61.3%	95.5%	82.9%
Clf. accuracy	73.5%	73.5%	66.5%	79.6%
Majority acc.	51.5%	66.3%	59.1%	78.2%
<i>Dream-7B</i>				
Compute saved	96.7%	36.0%	97.8%	61.4%
Clf. accuracy	72.4%	73.5%	70.0%	75.9%
Majority acc.	54.0%	61.5%	55.6%	74.5%

Table 2: Filtering simulation ($\tau = 0.80$). *Compute saved*: fraction of denoising steps avoided by skipping instances the probe predicts will fail (no output is produced for skipped instances). *Clf. accuracy*: fraction of instances for which the probe correctly predicts functional vs. non-functional. *Majority acc.*: trivial baseline that always predicts the majority class. Lift over majority is large for JSON (+19–22%) and MBPP (+7–15%) but only marginal for ARC (+1.4% on Dream).

softmax distribution, whose entropy or max probability could in principle serve the same role. We re-run the same filtering simulation using two such non-probe baselines (mean entropy and mean max-prob over masked positions in the generation region) and a logistic regressor trained on those scalar features (LR-entropy). Across all eight (dataset, model) settings, the probe outperforms the best raw signal in classification accuracy at $\tau = 0.80$ by 5–20 percentage points, and its peak per-step AUC exceeds LR-entropy by 0.05–0.18 on average (full tables in Appendix J). A second control trains a logistic regression directly on output-text features (length, format markers, and ARC predicted letter); this shallow probe is also outperformed by

the hidden-state probe in 7 of 8 settings with an average AUC gap of 0.12 and a maximum of 0.25 on GSM8K-Dream (Appendix K).

6 Discussion

Why does step-0 signal exist? At step 0, the DLM has performed a forward pass over the prompt with masked outputs, producing representations that correlate with eventual correctness. For structural tasks like JSON generation, this correlation is strong: the representation already contains signal predictive of correctness. For reasoning and code tasks, step-0 signal is weaker, and representations are iteratively refined through denoising. The DLM step-0 signal spans all generation positions simultaneously, arising from bidirectional attention over prompt and masked tokens. An AR baseline probe on Qwen-2.5-7B’s last prompt token achieves comparable AUC on most tasks (Appendix F), suggesting that much of the step-0 signal reflects prompt difficulty detectable by any model. On JSON schema, DLM step-0 AUC modestly exceeds the AR probe (0.78–0.80 vs. 0.74); on other tasks, the gap is negligible (Appendix F, Table 6). The distinctive advantage of DLMs lies not in step-0 signal strength but in the capacity to gain additional signal through denoising, a capacity absent in single-pass AR decoding entirely.

Relationship to temporal oscillation. Wang et al. (2025) observe that DLM outputs can oscillate during denoising: correct answers appear at intermediate steps but are overwritten later. Our probe measures whether hidden states *encode* correctness, not whether the current surface output is correct. On JSON schema, hidden-state signal is flat from step 0 while surface tokens are still being refined, suggesting that hidden-state signal predictive of correctness is established early and oscillation reflects surface-level instability rather than changes in the underlying representation.

Layer dynamics and training origin. LLaDA (trained from scratch) shows stable upper-layer localization (best layers L22–28 of 33) resembling AR models, while Dream (fine-tuned from Qwen-2.5) exhibits layer migration on simple tasks but not complex ones. On GSM8K, MBPP, and ARC, Dream’s best layers stay in L15–25 (of 29), tracking Qwen-2.5’s own probe best layers, which fall in L18–27 (of 28) on the same tasks (Appendix F). On JSON schema, however, Dream’s best layers

migrate down to L4–7 by step 32, deviating sharply from Qwen-2.5’s range. This pattern is consistent with the hypothesis that diffusion fine-tuning preserves the AR base’s upper-layer hierarchy on tasks that demand reasoning capacity, but unlocks alternative routing to lower layers when the task is structurally simple enough that the inherited hierarchy is no longer load-bearing. We caution that LLaDA’s block-based and Dream’s global linear schedules unmask different positions even at matched cumulative fractions, so step-axis comparisons (Figures 1, 2) reflect approximate denoising progress rather than identical model states.

7 Conclusion

We present the first probing study of DLM internals. Our central finding is that DLMs uniquely accumulate correctness signal through iterative denoising (a 0.08–0.11 AUC gain on reasoning tasks), a capacity absent in single-pass AR decoding. The step-0 hidden state already carries above-chance signal (AUC 0.61–0.80), but this is comparable to a probe on an AR model conditioned on the same prompt, indicating that step-0 signal reflects prompt difficulty rather than diffusion-specific computation. We additionally observe task-dependent emergence and distinct layer dynamics between LLaDA and Dream, and show that an offline filtering simulation based on per-step probe confidence avoids 36–98% of wasted compute and outperforms confidence heuristics derived from the model’s own softmax distribution. Length-stratified controls confirm the signal is not merely a length confound: AUC drops by 0.00–0.09 after length matching, with MBPP showing no drop at all.

Future work includes validating filtering in a real inference pipeline (with the probe forward pass costed in), scaling to larger DLMs, and exploring probe-guided adaptive compute allocation. Nonlinear or multi-layer probes may capture richer structure.

Limitations

Our probe is a simple logistic regression classifier with PCA-reduced features; nonlinear probes or multi-layer probes might reveal richer structure. The selective generation analysis is a simulation on states from full 128-step generation and does not account for inference overhead of running probes in real time. The AUC gain we observe through denoising (about 0.08–0.11 on reasoning tasks)

cannot be cleanly attributed to diffusion-specific computation versus generic iterative refinement: a single-pass AR model has no analogous comparison point, and we do not test multi-pass AR variants (e.g., self-consistency over multiple samples) that might also accumulate signal across passes.

We study only two models (LLaDA-8B, Dream-7B), both discrete masked diffusion LMs, and four tasks. Continuous diffusion LMs (e.g., Diffusion-LM, GENIE) operate on continuous embedding spaces with different denoising dynamics and are out of scope for this study; whether our findings extend to that family is an open question. Broader coverage across discrete masked architectures (e.g., MDLM, SEDD) and model scales is also needed. LLaDA (block-based) and Dream (global linear) reach approximately the same cumulative fraction of unmasked tokens at each checkpoint step (Figures 1, 2 are labeled accordingly), but the spatial distribution of unmasked tokens differs (LLaDA fills block-by-block, Dream picks most-confident globally), so cross-model comparisons should be read as approximate denoising-progress alignments rather than identical model states. Our interpretation of LLaDA vs. Dream layer dynamics in terms of training origin (§6) is correlational rather than causal: the two models also differ in pretraining data, tokenizer, and architecture, and a controlled fine-tuning ablation (e.g., diffusion fine-tuning the same AR base with and without lower-layer routing capacity) would be needed to causally establish the role of AR pretraining. A length control probe achieves higher AUC than the correctness probe in most settings (Appendix E); after length matching, correctness AUC drops modestly but remains well above chance. Prompt-length-matched probes similarly show only 0.02 average AUC drop (Appendix I). A shallow-feature probe trained jointly on output length, format markers, and ARC predicted letter is outperformed by the hidden-state probe in 7 of 8 settings (Appendix K); the exception is ARC-Dream where the shallow probe narrowly leads, consistent with the marginal probe lift over majority on ARC.

Probe selection (best layer and region) uses the same cross-validation splits as evaluation. A nested CV reanalysis (Appendix H) shows AUC drops by only 0.01 on average (max 0.07), confirming that the main patterns are not artifacts of selection bias. The filtering simulation in §5.3 treats the probe as free at inference time; deploying the same procedure live would incur additional cost

from a forward pass at each candidate exit step plus probe evaluation, and would require calibrating the threshold τ on a held-out development split rather than via cross-validation as we do here.

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A Generation and Probing Details

Denoising procedure. Both models run 128 denoising steps. LLaDA uses block-based unmasking: the generation region is divided into blocks of 32 tokens, and each block is denoised sequentially over $128/(\text{gen_length}/32)$ steps per block. Within

each block, the model selects the most confident tokens to unmask at each step. Dream uses a global linear schedule: at step i , the fraction of tokens to unmask is determined by a linearly decreasing timestep schedule from $t = 1$ to $t = \epsilon$, and the most confident masked tokens are unmasked globally.

Prompt construction. For JSON schema, we prepend a system prompt containing the target schema and append a JSON code-fence marker as a generation prefix. For GSM8K, the system prompt instructs the model to solve step-by-step and end with ##### followed by the numeric answer. Both models use their respective chat templates. A complete prompt example for each task is shown in Figure 3.

JSON schema (LLaDA chat template)

```
[BOS] system\n You are a helpful assistant
that answers in JSON. Here's the JSON schema
you must adhere to:\n <schema>\n {schema} \n
</schema>\n user\n {user input}\n assistant\n
""json\n
```

GSM8K (Dream chat template)

```
[BOS] im_start system\n Solve the math problem
step by step. End your answer with #####
followed by the final numeric answer. im_end\n
im_start user\n {question} im_end\n im_start
assistant\n
```

Figure 3: Schematic prompt examples after each model’s chat template is applied (special tokens shown as plain words for readability; the actual tokens used are model-specific control tokens). The generation region (128–512 masked tokens) is appended after the final assistant marker. Both models share the same system/user content; only the wrapping tokens differ.

Hidden state extraction. At each of the 7 checkpoint steps $\{0, 1, 4, 16, 32, 64, 127\}$, we run a forward pass with `output_hidden_states=True` and extract the full hidden state tensor across all layers. The generation region is partitioned into 4 equal-length position regions, and we mean-pool within each region, producing a feature vector of dimension d (4096 for LLaDA, 3584 for Dream) per layer per region per instance.

Probe pipeline. For each (step, layer, region) configuration, we construct the feature matrix $X \in \mathbb{R}^{n \times d}$ and apply: (1) StandardScaler (per-feature zero-mean and unit-variance normalization, with mean and variance fitted on the training fold and applied to the held-out fold), (2) PCA reduction to 64

components (also fitted on the training fold), (3) logistic regression with L2 regularization ($C = 1.0$, LBFGS solver, max 1000 iterations). We use 5-fold stratified cross-validation (random state 42) and report the mean AUC across folds.

Computational resources. All generation and feature extraction use NVIDIA A100 (80GB) GPUs on Modal, with 8 parallel A100s per experiment. Models are loaded in bfloat16 precision. Probe training runs on CPU. Total compute: approximately 30 A100-hours across all experiments.

B Additional Heatmaps

Figure 4 extends the heatmaps to MBPP and ARC. Per-model best-layer ranges (LLaDA MBPP L15–31; ARC L18–28 for both models) confirm that the JSON-schema downward migration in §5.2 does not occur on these tasks. ARC’s heatmaps are visibly washed out (most cells ≤ 0.65), consistent with the high majority-class baseline ($\geq 74\%$) limiting headroom.

C Best Layer per Step

Table 3: Best layer (by AUC) at each denoising step. Bold values highlight Dream’s migration to lower layers on JSON schema. Other tasks show no such migration.

Step	0	1	4	16	32	64	127
<i>JSON Schema</i>							
LLaDA	22	25	26	25	15	25	24
Dream	17	21	23	19	4	7	5
<i>GSM8K</i>							
LLaDA	24	24	24	25	28	24	25
Dream	23	21	21	21	21	21	19
<i>MBPP</i>							
LLaDA	15	26	25	24	27	23	31
Dream	15	15	24	18	19	25	19
<i>ARC-Challenge</i>							
LLaDA	26	25	25	24	25	25	23
Dream	20	20	19	19	19	19	19

D Selective Generation Details

The selective generation simulation works as follows. For each instance, we iterate through the checkpoint steps in order. At each step, we compute the probe’s confidence as $\max(p, 1-p)$ where p is the predicted probability of functional correctness. If the probe confidently predicts failure (confidence exceeds threshold τ and $p < 0.5$), we skip the remaining denoising steps for that instance, producing no output. If no checkpoint triggers skipping, the instance proceeds to full generation. Com-

pute savings are computed as $1 - \frac{t_{\text{skip}} + 1}{128}$, where t_{skip} is the step at which filtering occurs.

The main paper reports results at $\tau = 0.80$. Table 4 shows results across a range of thresholds for all configurations.

Table 4: Selective generation results across thresholds (%). “Sv” is compute saved by skipping instances predicted to fail. “Ac” is classification accuracy.

τ	JSON		GSM8K		MBPP		ARC	
	Sv	Ac	Sv	Ac	Sv	Ac	Sv	Ac
<i>LLaDA-8B</i>								
0.60	99.2	64.0	83.4	67.2	99.1	63.4	98.6	78.1
0.70	99.0	69.5	74.2	70.3	98.2	65.8	94.1	79.4
0.80	98.4	73.5	61.3	73.5	95.5	66.5	82.9	79.6
0.90	93.3	75.0	34.1	74.2	90.8	68.1	57.0	79.9
<i>Dream-7B</i>								
0.60	98.8	63.2	85.3	56.9	99.1	72.0	97.2	74.8
0.70	97.9	69.1	66.3	62.5	98.8	70.0	86.1	75.7
0.80	96.7	72.4	36.0	73.5	97.8	70.0	61.4	75.9
0.90	87.3	74.6	6.2	74.8	94.9	73.5	24.2	76.8

E Length Control Probe

We train a control probe (same pipeline) to predict binary output length (above/below median) instead of correctness. To control for the confound, we re-train the correctness probe on a length-matched subsample where functional and non-functional instances have identical length distributions (binned matching, 10 quantile bins).

Table 5: Best AUC for correctness, length control, and length-matched correctness probes. After controlling for length, correctness AUC drops modestly but remains well above chance.

	Corr.	Length	Corr. (matched)
<i>JSON Schema</i>			
LLaDA	0.85	0.98	0.81
Dream	0.84	0.99	0.75
<i>GSM8K</i>			
LLaDA	0.79	0.86	0.75
Dream	0.82	0.87	0.80
<i>MBPP</i>			
LLaDA	0.84	0.78	0.84
Dream	0.87	0.80	0.87
<i>ARC</i>			
LLaDA	0.75	0.96	0.75
Dream	0.71	0.98	0.72

F AR Baseline Probe

We probe Qwen-2.5-7B-Instruct (Dream’s AR base model) using the last prompt token hidden state. Sub-A predicts DLM labels; Sub-B uses Qwen’s own greedy outputs.

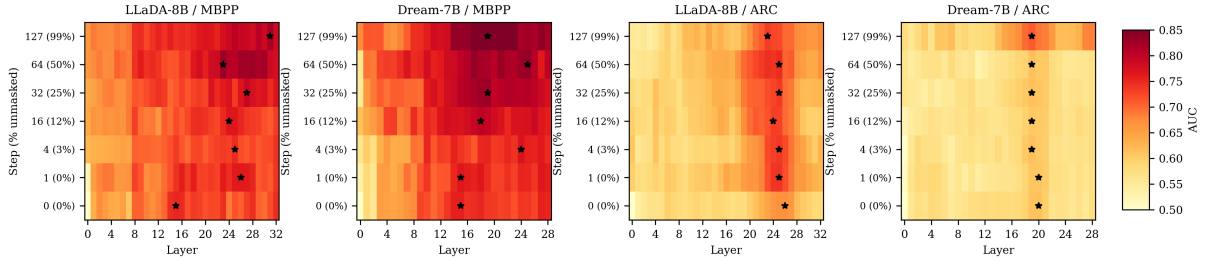


Figure 4: AUC heatmaps for MBPP and ARC-Challenge. Both show gradual emergence similar to GSM8K. ARC has the weakest overall signal ($\text{AUC} \leq 0.75$).

Table 6: AR vs. DLM step-0 probe AUC.

	JSON	GSM8K	MBPP	ARC
Qwen Sub-A (DLM)	0.74	0.69	0.76	0.67
Qwen Sub-B (AR)	0.77	0.77	0.69	0.74
LLaDA step-0	0.78	0.71	0.76	0.67
Dream step-0	0.80	0.71	0.76	0.61

G Spatial Pooling Ablation

We pool the 4 extracted regions into 1 (global average), 2 (pairwise average), or 4 (concatenation) groups and re-run the probe.

Table 7: Best AUC by number of position regions. Spread is less than 0.03 within each configuration.

	1 region	2 regions	4 regions
<i>JSON Schema</i>			
LLaDA	0.848	0.854	0.840
Dream	0.846	0.828	0.847
<i>GSM8K</i>			
LLaDA	0.784	0.788	0.784
Dream	0.818	0.814	0.812
<i>MBPP</i>			
LLaDA	0.842	0.860	0.834
Dream	0.864	0.868	0.856
<i>ARC-Challenge</i>			
LLaDA	0.747	0.740	0.744
Dream	0.710	0.711	0.716

H Nested Cross-Validation

To verify that probe selection does not inflate reported AUC, we rerun all experiments with nested cross-validation: an inner 3-fold CV on training data selects the best (layer, region), and the outer 5-fold evaluates on held-out data.

Table 8: AUC drop from nested vs. standard CV. Average drop is 0.01; all qualitative patterns are preserved.

	Avg Drop	Max Drop
<i>JSON Schema</i>		
LLaDA	0.005	0.050
Dream	0.035	0.072
<i>GSM8K</i>		
LLaDA	0.009	0.019
Dream	0.008	0.014
<i>MBPP</i>		
LLaDA	0.025	0.053
Dream	0.021	0.038
<i>ARC-Challenge</i>		
LLaDA	0.002	0.015
Dream	0.005	0.031

I Prompt Length Control

We re-train correctness probes on subsamples where functional and non-functional instances have matched prompt length distributions (binned matching, 10 quantile bins).

Table 9: AUC after controlling for prompt length. Average drop is 0.02.

	Original	Prompt-matched	Drop
<i>JSON Schema</i>			
LLaDA	0.80	0.75	0.052
Dream	0.82	0.80	0.022
<i>GSM8K</i>			
LLaDA	0.78	0.77	0.015
Dream	0.82	0.80	0.017
<i>MBPP</i>			
LLaDA	0.81	0.86	-0.046
Dream	0.88	0.85	0.028
<i>ARC-Challenge</i>			
LLaDA	0.73	0.70	0.030
Dream	0.71	0.68	0.029

J Entropy / Confidence Baseline

For each step we extract masked positions in the generation region and compute (a) mean entropy and (b) mean max-probability, pooled into the same 4 spatial regions as the probe. We evaluate three confidence signals: **Raw -entropy** ($-H$ pooled over regions), **Raw maxprob** (mean max-prob),

and **LR-entropy** (logistic regression on the 8-dim feature vector, mirroring the probe’s CV pipeline).

Table 10 reports the best per-step AUC across checkpoints for each method. The probe outperforms the strongest baseline in every (dataset, model) setting, with an average gap of 0.097 AUC and a maximum gap of 0.179 on GSM8K-Dream. Table 11 reports filtering performance at $\tau = 0.80$ matching Table 2; the probe achieves higher classification accuracy than every confidence baseline on 7 of 8 settings (the exception is JSON-Dream, where LR-entropy reaches 78.3% but at less than half the compute saved).

Table 10: Best per-step AUC across checkpoints. Probe is the hidden-state logistic regression from §5.3; the other three columns use only the model’s softmax distribution at each step.

	Probe	LR-ent	−H	maxprob
<i>JSON Schema</i>				
LLaDA	0.846	0.790	0.738	0.746
Dream	0.827	0.815	0.785	0.798
<i>GSM8K</i>				
LLaDA	0.785	0.701	0.675	0.667
Dream	0.818	0.639	0.646	0.646
<i>MBPP</i>				
LLaDA	0.810	0.730	0.724	0.725
Dream	0.879	0.776	0.749	0.744
<i>ARC-Challenge</i>				
LLaDA	0.745	0.611	0.623	0.610
Dream	0.713	0.588	0.592	0.580

Table 11: Filtering simulation at $\tau = 0.80$, formatted as compute-saved% / accuracy%. Probe column matches Table 2.

	Probe	LR-ent	−H	maxprob
<i>JSON Schema</i>				
LLaDA	98.4 / 73.5	41.0 / 61.0	80.3 / 52.2	82.1 / 54.4
Dream	96.7 / 72.4	43.1 / 78.3	85.8 / 50.0	83.1 / 52.2
<i>GSM8K</i>				
LLaDA	61.3 / 73.5	36.7 / 68.8	59.3 / 68.5	66.9 / 66.6
Dream	36.0 / 73.5	6.3 / 62.2	30.5 / 62.2	31.0 / 62.2
<i>MBPP</i>				
LLaDA	95.5 / 66.5	34.8 / 60.7	78.3 / 48.2	82.4 / 52.9
Dream	97.8 / 70.0	29.3 / 67.3	87.0 / 52.9	86.6 / 52.1
<i>ARC-Challenge</i>				
LLaDA	82.9 / 79.6	77.1 / 78.0	93.4 / 70.4	96.4 / 68.4
Dream	61.4 / 75.9	43.5 / 74.3	49.7 / 60.8	67.3 / 51.8

K Shallow-Feature Control

To test whether the hidden-state probe merely picks up surface cues, we extract per-task textual features from each generation and train a 5-fold cross-validated logistic regression on them ("shallow probe"). Features cover output length (charac-

ter/line count), format markers (presence of expected delimiters such as ####, “python, code fences, JSON parseability, def), and, for ARC, one-hot indicators of the predicted answer letter (A/B/C/D). We compare the shallow probe’s AUC to the best per-step hidden-state probe.

Table 12 summarizes the comparison. The hidden-state probe outperforms the shallow probe in 7 of 8 (dataset, model) settings, with an average gap of 0.12 AUC and a maximum gap of 0.25 on GSM8K-Dream. The single exception is ARC-Dream (−0.01), consistent with the marginal lift over majority baseline already noted for ARC in Table 2. On ARC, restricting the shallow probe to the predicted-letter one-hot alone yields AUC 0.640 for LLaDA and 0.681 for Dream, indicating that the answer letter is a partial but not dominant component of the shallow signal.

Table 12: Hidden-state probe vs. shallow-feature probe (logistic regression on output length, format markers, and ARC predicted letter). Probe column is the best per-step AUC across checkpoints; shallow is mean AUC over 5 folds.

	Probe	Shallow	Gap
<i>JSON Schema</i>			
LLaDA	0.846	0.693	+0.15
Dream	0.827	0.738	+0.09
<i>GSM8K</i>			
LLaDA	0.785	0.685	+0.10
Dream	0.818	0.567	+0.25
<i>MBPP</i>			
LLaDA	0.810	0.678	+0.13
Dream	0.879	0.673	+0.21
<i>ARC-Challenge</i>			
LLaDA	0.745	0.679	+0.07
Dream	0.713	0.725	−0.01